

# Dr.-Ing. Danish IQBAL

Computational Engineer

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📍 Essen, Germany



[m-d-iqbal.github.io](https://m-d-iqbal.github.io) [linkedin.com/in/drdenishiqbal](https://linkedin.com/in/drdenishiqbal) [scholar.google.de/citations?user=cxAW-3MAAAAJ](https://scholar.google.de/citations?user=cxAW-3MAAAAJ) [github.com/M-D-Iqbal](https://github.com/M-D-Iqbal)

## PROFILE

Dr.-Ing. in Computational Mechanics with seven years of research in numerical methods, multi-physics simulation, and fracture mechanics. Hands-on experience developing custom numerical methods and using commercial FEM simulation software, four peer-reviewed journal publications with over 135 citations, and international research collaboration. Complemented by industry experience in production software development, workflow automation, and applied AI integration.

## PROFESSIONAL EXPERIENCE

07/2025 - present

### Software Engineer

CleverBuy GmbH, Krefeld, Germany

- Architected and shipped a pricing platform for 30,000+ refurbished-electronics SKUs on Refurbed and Back Market, handling both sell-side listings and buyback offers, with a monitoring and override dashboard that cut the operations lead's manual pricing work by 80%.
- Designed the PostgreSQL service-layer schema on top of the company's JTL-Wawi ERP and built a 24/7 priority-based asynchronous pipeline that reprices the catalogue under explicit margin and grade-tier rules.
- Integrated Refurbed and Back Market REST APIs and the Anthropic Claude API for LLM-supported grade and pricing decisions, with automated SKU onboarding and product-image processing for the company storefront.

08/2017 - 03/2024

### Research Assistant

University of Duisburg-Essen, Essen, Germany

- Developed a novel polygon-based SBFEM formulation for transient thermo-elastic fracture in functionally graded materials, implemented in object-oriented MATLAB and validated against commercial FEM software.
- Published four peer-reviewed journal papers with over 135 citations in top computational- and fracture-mechanics journals; peer reviewer for an international journal.
- Designed and delivered a master's-level Structural Dynamics course in ANSYS on a university e-learning grant, including a custom interactive simulation lab for hands-on structural-dynamics teaching.
- Taught a master's-level Computer Languages for Engineers course in object-oriented Fortran and C++ with numerical-methods term projects.
- Supervised three master's and one bachelor's thesis in computational mechanics and CFD.

01/2020 - 03/2020

### Visiting Researcher

Federation University, Ballarat, Australia

- Australia-Germany research cooperation (Federation University, DAAD) on polygon SBFEM for coupled thermoelastic fracture in functionally graded materials.

PROFESSIONAL EXPERIENCE

|                   |                                                                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01/2017 - 03/2017 | <b>Student Research Assistant</b><br>University of Duisburg-Essen, Essen, Germany <ul style="list-style-type: none"><li>Implemented a polygon-based SBFEM in Fortran and integrated it into the open-source FE code FEAppv.</li></ul> |
| 06/2015 - 10/2015 | <b>Seasonal Worker (Assembly Line)</b><br>Mercedes-Benz Group AG, Düsseldorf, Germany <ul style="list-style-type: none"><li>Spot welding with jigs and automotive assembly-line work.</li></ul>                                       |
| 10/2014 - 12/2014 | <b>Seasonal Worker (Assembly Line)</b><br>Mercedes-Benz Group AG, Rastatt, Germany <ul style="list-style-type: none"><li>Production tasks on automotive assembly line.</li></ul>                                                      |
| 12/2013 - 12/2013 | <b>Seasonal Worker (Logistics)</b><br>Amazon Deutschland Services GmbH, Rheinberg, Germany <ul style="list-style-type: none"><li>Warehouse scanning and packaging preparation.</li></ul>                                              |
| 06/2011 - 07/2011 | <b>Internship</b><br>Karachi Tools, Dies & Moulds Center, Karachi, Pakistan <ul style="list-style-type: none"><li>Practical training in CAD/CAM and nitriding heat treatment.</li></ul>                                               |

CORE COMPETENCIES

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|----------------------------|------------------------|---------------------------|
| • Computational Mechanics  | • Numerical Methods    | • Finite Element Analysis |
| • Multi-Physics Simulation | • Fracture Mechanics   | • Software Development    |
| • Applied AI Integration   | • Data Pipelines & ETL | • Project Management      |

SKILLS

|                    |              |              |           |                      |
|--------------------|--------------|--------------|-----------|----------------------|
| Numerical Methods: | • FEM        | • SBFEM      | • XFEM    | • CFD                |
| Simulation Tools:  | • ANSYS      | • Abaqus     | • Gmsh    | • ParaView           |
| Programming:       | • MATLAB     | • Python     | • Fortran | • C++                |
| Backend & Data:    | • Django     | • PostgreSQL | • Celery  | • REST APIs          |
| Applied AI:        | • Claude API | • Gemini     | • Groq    | • Prompt engineering |

EDUCATION

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|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08/2017 - 07/2024 | <b>Ph.D. (Dr.-Ing.) Computational Mechanics</b><br>University of Duisburg-Essen, Germany <ul style="list-style-type: none"><li>Thesis: Development of scaled boundary polygon elements for coupled thermoelastic fracture modeling.</li><li>Grade: Magna cum laude (1.0)</li></ul> |
| 10/2013 - 07/2017 | <b>M.Sc. Computational Mechanics</b><br>University of Duisburg-Essen, Germany <ul style="list-style-type: none"><li>Thesis: Development of polygon elements based on the scaled boundary finite element method.</li><li>Grade: Good (2.0)</li></ul>                                |

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## EDUCATION

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01/2009 - 12/2012

B.E. Automotive Engineering

NED University of Engineering & Technology, Karachi, Pakistan

- Thesis: Design and analysis of motorcycle oil pump.
- Grade: First Division

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## PERSONAL PROJECTS

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sahab.pk — AI-Powered Ledger SaaS

Founder & architect — live alpha

- Multi-tenant SaaS for retailers in Pakistan: shopkeepers chat in Urdu or English; an LLM extracts structured events into a double-entry ledger with row-level tenant isolation.

Free-Body Diagram Toolkit — Teaching Aid

Developer & maintainer — open source

- MATLAB code used in own Ph.D. thesis and in graduate Structural Dynamics tutorials, assignments, and exams at the University of Duisburg-Essen.

Formula Student Race Car — Carbon-Fiber Monocoque

Undergraduate team project — NED University, Karachi

- Designed to Formula Student Spain competition specification: carbon-fiber monocoque in Siemens NX with CFRP/honeycomb sandwich construction, panel strength testing passed; prior ANSYS structural analysis on the predecessor space-frame chassis.

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## PUBLICATIONS

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### Peer-Reviewed Journal Papers

*M. D. Iqbal*, et al. "Transient thermoelastic fracture analysis of functionally graded materials using the scaled boundary finite element method." *Theoretical and Applied Fracture Mechanics*, vol. 127, Oct. 2023, p. 104056.

*M. D. Iqbal*, et al. "Thermoelastic fracture analysis of functionally graded materials using the scaled boundary finite element method." *Engineering Fracture Mechanics*, vol. 264, Apr. 2022, p. 108305.

*M. D. Iqbal*, et al. "Development of the scaled boundary finite element method for crack propagation modeling of elastic solids subjected to coupled thermo-mechanical loads." *Computer Methods in Applied Mechanics and Engineering*, vol. 387, Dec. 2021, p. 114106.

E.T. Ooi, *M. D. Iqbal*, et al. "A polygon scaled boundary finite element formulation for transient coupled thermoelastic fracture problems." *Engineering Fracture Mechanics*, vol. 240, Dec. 2020, p. 107300.

### Peer-Reviewed Conference Papers

C. Birk, R. Assaf, *M. D. Iqbal*, H. Gravenkamp. "Numerische Strukturmodellierung mit Polygon-Elementen." *Baustatik-Baupraxis* 15, 2024.

*M. D. Iqbal*, et al. "Scaled boundary finite element modelling of thermally induced crack propagation." *Proceedings in Applied Mathematics and Mechanics (PAMM)* 19, 2019.

### Conference Presentations

*M. D. Iqbal*, et al. "Scaled boundary polygon element formulation for coupled thermoelastic fracture in FGMs." *GACM Colloquium on Computational Mechanics*, 2022.

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## PUBLICATIONS

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M. D. Iqbal, et al. "Polygon SBFEM for transient coupled thermoelastic fracture." 14th World Congress on Computational Mechanics (WCCM XIV), 2020 (virtual).

M. D. Iqbal, et al. "Scaled boundary finite element modelling of thermally induced crack propagation." 90th GAMM Annual Meeting, 2019.

M. D. Iqbal, C. Birk. "Numerical modelling of thermally induced crack propagation in functionally graded materials using the scaled boundary finite element method." Forschungskolloquium Baustatik-Baupraxis, Grasellenbach, 2018.

## Peer Review

Reviewer, *International Journal of Non-Linear Mechanics* – "Thermally nonlinear analysis of propagating cracks under generalized thermal shock".

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## LANGUAGES

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Urdu (native)

English (C2)

German (fluent, B2)

Essen, 24 May 2026

